Principled Pronouns

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Introduction

Meaning depends on context.

Certain elements *grammatically* encode a dependency on context:

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- **Pronouns** refer to entities in the context.

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- **Pronouns** refer to entities in the context.

(1) The new candidate is great. I hope she wins.

- **Presuppositions** demand that certain information be known in context.

(2) Jane was here this morning. I'm wondering if she's still here.

The different constituents of an utterance don't all access the same "*context*".

(3) **Pronouns**

a. If there was a phone-book, it would be in the cabinet.

b. If there was a phone-book, our quality of life would improve. # It is in the cabinet.

(4) **Presuppositions**

a. If Jane was here at some point, she still is.

b. If Jane was here at some point, then we're good.
She's still here by the way.

Accessibility conditions for pronouns

When can a pronoun refer to a given antecedent?

Presupposition satisfaction for presuppositions

When is a presupposition met?

(5) a. There was a phone-book and it was in the cabinet.

b. If there was a phone-book, it would be in the cabinet.

c. There might be a phone-book ; it would be in the cabinet.

(5) a. There was a phone-book and it was in the cabinet.

b. If there was a phone-book, it would be in the cabinet.

c. There might be a phone-book ; it would be in the cabinet.

(6) a. # There wasn't a phone-book and it is in the cabinet.

b. # If there never was a phone-book, it was in the cabinet.

c. # If there was a phone-book, our quality of life would improve. It is in the cabinet.

You can only refer to the phone book if the existence of the phone-book can be taken for granted when the pronoun is used?

Claims

- A pronoun can refer to its antecedent if and only if its referent can be presupposed to exist.

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- A pronoun can refer to its antecedent if and only if its referent can be presupposed to exist.
- It may seem obvious but the generalization is typically not validated in theories of pronoun accessibility, typically Dynamic Semantics.
- I propose a system in which pronouns are interpretable as soon as the existence presupposition is met.

Benefits

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- It reduces pronoun accessibility to presupposition satisfaction.
- It does not require significantly altering the semantics.

Precursor: Dynamic Semantics

Dynamic Semantics is a unified framework for presuppositions and pronouns.
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Founding assumption of Dynamic Semantics

The meaning of an expression *is* its effect on context.

(7) ✓ A photographer$_{12}$ came in and they$_{12}$ sat down.

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(8) a. CP_1 and CP_2

b. *update the context with* $[\![CP_1]\!]$,
then update the context with $[\![CP_2]\!]$

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(9) a. a photographer$_{12}$

b. *add a photographer at index 12 of the context*

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(9) a. a photographer$_{12}$

b. *add a photographer at index 12 of the context*

(10) a. they$_{12}$

b. *fetch the entity at index 12 from the context*

(11) ✓ Jane was here this morning and she's still here.

(12) a. CP_1 and CP_2

b. *update the context with* $[\![CP_1]\!]$,
then update the context with $[\![CP_2]\!]$

(13) a. Jane was here this morning.

b. *add to the common ground the fact that Jane was here this morning*

(14) a. She's still here.

b. *crash if the fact that Jane was here before isn't part of the common ground*

(15) a. # She$_{12}$ came in and a photographer$_{12}$ sat down.

b. # She's still here and Jane was here this morning.

(16) a. CP_1 and CP_2

b. *update the context with* $[\![CP_1]\!]$,
then update the context with $[\![CP_2]\!]$

Two problems:

1. **The explanatoriness problem:** [Soames, 1989]
   the space of possible dynamic meanings is too rich.
2. **The under-generation problem:**
   [Groenendijk and Stokhof, 1991]
   certain felicitous pronouns are predicted not to be.

The explanatoriness problem

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b. update the context with $[\![CP_1]\!]$,
then update the context with $[\![CP_2]\!]$

(18) a. CP_1 *wand* CP_2

b. update the context with $[\![CP_2]\!]$,
then update the context with $[\![CP_1]\!]$

The explanatoriness problem

(19) a. They came in *wand* a photographer sat down.

b. # A photographer came in *wand* they sat down.

In a nutshell: *wand* would only allow cataphora.

(20) a. Jane is still here *wand* she was here before.

b. # Jane was here before *wand* she's still here.

In a nutshell: *wand* would only allow right-to-left presupposition satisfaction.

The explanatoriness problem

Cross-linguistic: is *wand* ever attested? if not, why?

Acquisition: do we see a stage where children have acquired the truth-conditional effect of *and*, but not its "*direction*"? if not, why?

A theoretical desideratum:
A theory with the following architecture would be desirable:

syntax + truth-conditional semantics $\Rightarrow$ context effects

For the case of presuppositions, alternative theories of presuppositions have met the explanatoriness challenge [Schlenker, 2009, George, 2008, Rothschild, 2011].

At the end of this talk, we'll present a system that extends this to pronouns [Elliott, 2020a, Mandelkern, 2020].

Two problems:

1. The explanatoriness problem:
   the space of possible dynamic meanings is too rich.
2. **The under-generation problem:**
   certain felicitous pronouns are predicted not to be.

Under-generation

Vanilla Dynamic Semantics [Groenendijk and Stokhof, 1991] does not predict the following sentences to be acceptable:

(21) Predicted #

a. Either there wasn't a phone-book or it was in the cabinet. (bathroom sentences)

b. It's not true that there isn't a phone-book. It's hidden in the cabinet. (double negation)

Under-generation

(22) a. There was a phone-book and it was in the cabinet.

b. Either there wasn't a phone-book or it was in the cabinet.

c. If there was a phone-book, it was in the cabinet.

(23) a. # There wasn't a phone-book and it is in the cabinet.

b. # Either there was a phone-book or it was in the cabinet.

c. # If there was a phone-book, our quality of life would improve. It is in the cabinet.

Under-generation

Informal generalization

In a linguistic context "*...there was a phone-book ...it*", *it* may co-refer with *a phone-book* if and only if the existence of a phone-book can be presupposed at the point when the pronoun is used.

Under-generation

Informal generalization

In a linguistic context "*...there was a phone-book ...it*", *it* may co-refer with *a phone-book* if and only if the existence of a phone-book can be presupposed at the point when the pronoun is used.

Full generalization

In a context "*...a/some NP VP ...it*", *it* may co-refer with *a NP* if and only if the presupposition that there is a NP that VP's is satisfied.

(Cf also [Mandelkern and Rothschild, 2019])

(24) Predicted #

a. Either there wasn't a phone-book or it was in the cabinet. (bathroom sentences)

b. It's not true that there isn't a phone-book. It's hidden in the cabinet. (double negation)

Dynamic Semantics only validates one direction of the generalization: when *it* is used, the existence of the phone-book can be presupposed.

Fixes [Elliott, 2020b, Krahmer and Muskens, 1995] exist, typically at two costs:

- …richer spaces of meaning.
- …leave other empirical gaps open [Hofmann, 2019].

Proposal

Informal generalization

In a context "…*there was a phone-book* …*it*", *it* may co-refer with *a phone-book* if and only if the existence of a phone-book can be presupposed when the pronoun is used.

Full generalization

In a context "…*a NP VP* …*it*", *it* may co-refer with *a NP* if and only if the presupposition that there is a NP that VP's is satisfied.

- Pronouns carry an existence presupposition.
- As soon as the presupposition is met, the pronoun will be interpretable.
- Building on any theory of presuppositions satisfaction, including "*explanatory*" theories [Schlenker, 2009, Fox, 2013, George, 2008, Rothschild, 2011].

Pronoun structure

(25) $[\![\text{it}_i]\!] = f_i(W_i)$

Where W and f are two parameters of interpretation such that:

- for every i, W_i is the witness set.
  (e.g. the set of phone-books)
- for every i, f_i is a choice function: $\forall S, f(S) \in S$
  (e.g. the phone-book we want to pick)

Pronoun structure

There are pragmatic principles constraining f_i and W_i. For now:
For any set, there is an i such that $W_i = S$

Rule for W

If there is a constituent "*[a NP]*$_i$ *VP*" in the neighboring linguistic context, then it has to be the case that $W_i = [\![\text{NP}]\!] \cap [\![\text{VP}]\!]$.

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Rule for W

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(26) There was a phone-book$_1$ ...
$\rightsquigarrow W_1 = [\![\text{phone-book}]\!] \cap [\![\text{there}]\!]$

Pronoun structure

There are pragmatic principles constraining f_i and W_i. For now:
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Rule for W

If there is a constituent "*[a NP]*$_i$ *VP*" in the neighboring linguistic context, then it has to be the case that $W_i = [\![\text{NP}]\!] \cap [\![\text{VP}]\!]$.

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$\rightsquigarrow W_1 = [\![\text{phone-book}]\!] \cap [\![\text{there}]\!]$

Provisional f_i

No constraint apart from it being a choice function.

Anaphora in conjunction

(27) a. There was a phone-book$_1$ and it$_1$ was in the cabinet.

b. $[\![\text{it}_1]\!] = f_1(W_1)$

Anaphora in conjunction

(27) a. There was a phone-book$_1$ and it$_1$ was in the cabinet.

b. $[\![\text{it}_1]\!] = f_1(W_1) \leftarrow$ choice functions can't apply to empty sets!

Anaphora in conjunction

(27) a. There was a phone-book$_1$ and it$_1$ was in the cabinet.

b. $[\![\text{it}_1]\!] = f_1(W_1) \leftarrow W_1$ should not be empty!

c. **Presupposition:** $W_1 = [\![\text{phone-book}]\!] \cap [\![\text{there}]\!] \neq \varnothing$

Anaphora in conjunction

(28) There is a phone-book$_1$ and $\underbrace{\text{it}_1 \text{ is in the cabinet}}_{\text{pres: there is a phone-book}}$.

From theories of presuppositions satisfaction:
p and $\underline{p}q$ does not carry a presupposition.

(28) There is a phone-book$_1$ and it$_1$ is in the cabinet.

pres: $\varnothing$

From theories of presuppositions satisfaction:
p and $\underline{p}q$ does not carry a presupposition.

(29)# It$_1$ is in the cabinet and there is a phone-book$_1$.
pres: there is a phone-book

From theories of presupposition satisfaction:
The presupposition of *p'p and q* is *p'*
[Schlenker, 2009, Fox, 2013, George, 2008].

(29)# It$_1$ is in the cabinet and there is a phone-book$_1$.

pres: there is a phone-book

From theories of presupposition satisfaction:
The presupposition of *p'p and q* is *p'*
[Schlenker, 2009, Fox, 2013, George, 2008].

Cataphora in conjunction

The ban on cataphora is related to the impossibility of right-to-left presupposition satisfaction [Mandelkern et al., 2020].

(30) a. There is a phone-book$_1$ and it$_1$ is in the cabinet.

b. There was a phone-book and there still is a phone-book.

(31) a. # It$_1$ is in the cabinet and there is a phone-book$_1$.

b. # There still is a phone-book and there was a phone-book.

Exceptions to the ban on cataphora

But there are exceptions to the ban on right-to-left presupposition satisfaction:

(32) Jane's no longer in Norway but she was there at some point.

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But there are exceptions to the ban on right-to-left presupposition satisfaction:

(32) Jane's no longer in Norway but she was there at some point.

NB: *but* is not *wand*

(33) Jane was in Norway at some point but she's no longer there.

Exceptions to the ban on cataphora

But there are exceptions to the ban on right-to-left presupposition satisfaction:

(32) Jane's no longer in Norway but she was there at some point.

Such configurations also license cataphora:

(33) She$_1$ isn't in Norway now but one of my students$_1$ studied there.

Bathroom sentences

(34) Either there wasn't a phone-book$_1$
or <u>it$_1$ was in the cabinet</u> .
pres.: there was a phone-book

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From theories of presupposition satisfaction:
either p or <u>not p</u>q does not carry a presupposition.

Bathroom sentences

These cases raise no particular issues in this theory.

Parallels pronouns/presuppositions

(35) a. There was a phone-book and it was in the cabinet.

b. There was a phone-book and there still is.

Parallels pronouns/presuppositions

(35) a. There was a phone-book and it was in the cabinet.

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b. If there was a phone-book, there still is.

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(35) a. There was a phone-book and it was in the cabinet.

b. There was a phone-book and there still is.

(36) a. If there was a phone-book, it was in the cabinet.

b. If there was a phone-book, there still is.

(37) a. There might have been a phone-book and it would have been in the cabinet.

b. There might have been a phone-book and there would still be one.

Parallels pronouns/presuppositions

(35) a. There was a phone-book and it was in the cabinet.

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(36) a. If there was a phone-book, it was in the cabinet.

b. If there was a phone-book, there still is.

(37) a. There might have been a phone-book and it would have been in the cabinet.

b. There might have been a phone-book and there would still be one.

(38) a. Either there wasn't a phone-book or it was in the cabinet.

b. Either there wasn't a phone-book or there still is.

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(39) a. # There wasn't a phone-book and it is in the cabinet.

b. # There wasn't a phone-book and there still is.

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Parallels pronouns/presuppositions

(39) a.# There wasn't a phone-book and it is in the cabinet.

b.# There wasn't a phone-book and there still is.

(40) a.# There might have been a phone-book and it was in the cabinet.

b.# There might have been a phone-book and there still is.

(41) a.# Either there was a phone-book or it was in the cabinet.

b.# Either there was a phone-book or there still is.

Accessibility via pragmatic inferences

(42) Bill believes that I don't have a sister$_1$.
$\rightsquigarrow$ *I have a sister.*

Maximize Presupposition: [Heim, 1991, Chemla, 2008]
When speaker is likely well-informed about *p*, "*Bill believes p*" conveys *p* is false.

A presupposition can be satisfied by such a pragmatic inference:

(43) Bill believes that I don't have a sister$_1$.
Yet, he's met my sister$_1$ several times.

So can pronouns:

(44) Bill believes I don't have a sister$_1$.
Yet, he's met her$_1$ several times.

The inference is defeasible; when defeated, the pronoun becomes unavailable:

(45) Bill believes, <u>and he's correct,</u> that I don't have a sister$_1$.
Yet, he's met her$_1$ several times.

Precedents and inspirations

This theory finds parallels in:

- **E-type** theories assume that pronouns are definite descriptions: *it = the phone-book that there was*.
  - Unamended, they predict problematic uniqueness inferences and the question of the *formal link*.
  - This proposal drops the assumption of uniqueness and gives an answer to the formal link problem.

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- **E-type** theories assume that pronouns are definite descriptions: *it = the phone-book that there was*.
  - Unamended, they predict problematic uniqueness inferences and the question of the *formal link*.
  - This proposal drops the assumption of uniqueness and gives an answer to the formal link problem.
- [Hofmann, 2019]'s ICDRT:
  - Indefinites introduce individual concepts in a dynamic system and carry presuppositions.
  - This leads to a system where a pronoun's accessibility is dictated by both dynamic meanings and presupposition satisfaction.

Interim summary

- Pronouns' interpretation contains two parts: the witness set W and a choice function f.
- A general principle specifies the nature of W.
- Pronouns are interpretable iff W is not empty.
- Building on a theory of presupposition satisfaction, predicts:
  - Simple conjunctions.
  - Ban on cataphora.
  - Exceptions to the ban.
  - Bathroom sentences
  - Intrusion of pragmatic inferences in pronoun's accessibility conditions

Quantificational cases

(46) Every farmer owns a donkey$_1$.
Few of them feed it$_1$ oats.

(47) Every farmer who owns a donkey$_1$ feeds it$_1$ oats.

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Few of them feed it$_1$ oats.

(47) Every farmer who owns a donkey$_1$ feeds it$_1$ oats.

Why these cases have garnered a lot of interest:

- co-variation without c-command.
- the speaker has no particular donkey in mind.

(48) Every farmer owns a donkey$_1$.
Few of them feed it$_1$ oats.

a. Every farmer owns a donkey$_1$.

b. [every farmer] λx. [a donkey]$_1$ λy. x owns y

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If there is a constituent "*[a NP]$_i$ VP*" in the context, then $W_i = [\![\text{NP}]\!] \cap [\![\text{VP}]\!]$.

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If there is a constituent "*[a NP]$_i$ VP*" in the context, then $W_i(g) = [\![\text{NP}]\!]^g \cap [\![\text{VP}]\!]^g$.

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Rule for W

If there is a constituent "*[a NP]$_i$ VP*" in the context, then $W_i(g) = [\![\text{NP}]\!]^g \cap [\![\text{VP}]\!]^g$.

In this case: $W_1(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey} \\ g(x) \text{ owns } y \end{array} \right\}$

(49) a. Few of them feed it_1 oats.

b. Few of them $\lambda x.\ x$ feed it_1 oats.

c. $W_1(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey} \\ g(x) \text{ owns } y \end{array} \right\}$

(49) a. Few of them feed it_1 oats.

b. Few of them $\lambda x.$ $\underline{x \text{ feed } \text{it}_1 \text{ oats}}$

pres.: $W_1(g) \neq \varnothing$

c. $W_1(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey} \\ g(x) \text{ owns } y \end{array} \right\}$

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pres.: $g(x)$ owns a donkey

c. $W_1(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey} \\ g(x) \text{ owns } y \end{array} \right\}$

Universal presupposition projection out of *few*:

(50) a. Few of the farmers still have a donkey.
$\rightsquigarrow$ *all of them had a donkey*

b. Few of the farmers $\lambda x.$ $\underbrace{x \text{ still have a donkey}}_{\text{pres.: } g(x) \text{ had a donkey}}$.

c. $\underbrace{\text{Few of the farmers } \lambda x.\ x \text{ still have a donkey}}_{\text{pres.: all farmers had a donkey}}$.

(51) a. Every farmer owns a donkey$_1$ and few of them feed it$_1$ oats.

b. Few of them $\lambda x.$ $\underline{x \text{ feed it}_1 \text{ oats}}$

pres.: $g(x)$ owns a donkey

c. $\underline{\text{Few of them } \lambda x.\ x \text{ feed it}_1 \text{ oats}}$

pres.: all of them own a donkey

The same reasoning applies to the donkey sentence.

(52) a. Every farmer who owns a donkey$_1$

$\lambda x.\ \underbrace{x \text{ feeds it}_1 \text{ oats}}_{\text{pres.: } W_1(g) \neq \varnothing}$

b. $W_1(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey} \\ g(x) \text{ owns } y \end{array} \right\}$

The same reasoning applies to the donkey sentence.

(52) a. Every farmer who owns a donkey$_1$

$\lambda x.$ $\underline{x \text{ feeds it}_1 \text{ oats}}$

pres.: $g(x)$ owns a donkey

b. $W_1(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey} \\ g(x) \text{ owns } y \end{array} \right\}$

The set W_i may depend on the value of certain binders.

What's next

- **Determining** W**:** what referent does a DP make available?
- **Determining** f**:** what happens when a DP has multiple witnesses?

Determining W

Rule for W

If there is a constituent "*[a NP]*$_i$ *VP*" in the context, then $W_i(g) = [\![\text{NP}]\!]^g \cap [\![\text{VP}]\!]^g$.

How does a language know what *a NP* is?
$\rightsquigarrow$ *this can't be a general principle of interpretation.*

There must be a general recipe for determining whether and what (quantifier) DPs can antecede pronouns:

(53) a. Some donkey brayed and then it ran away.

b. Every donkey brayed and then they ran away.

(54) a. Some donkey brayed and then it ran away.

b. Every donkey brayed and then they ran away.

Intuition

A quantified statement allows one to refer to the smallest individual which served to make the statement true.

Adjacent ideas: *minimal situations* [Kratzer, 2002, Elbourne, 2005], *exact truth-makers* [Fine, 2017], ...

Intuition

A quantified statement allows one to refer to the smallest individual which served to make the statement true.

(55) a. Some donkey brayed.

b. Some donkey brayed and is a part of X.

(56) a. **Witnesses:** any X containing some braying donkeys

b. **Smallest witnesses:** x where x is a braying donkey

Intuition

A quantified statement allows one to refer to the smallest individual which served to make the statement true.

(57) a. Every donkey brayed.

b. Every donkey brayed and is a part of X.

(58) a. **Witnesses:** any X containing all the braying donkeys

b. **Smallest witness-$\varnothing$:** the plurality of braying donkeys

With some amendments and simplifications [Szabolcsi, 2012]:

Witness set

X is a witness for $Q(A)(B)$, if $Q(A)(B \cap \{x \mid x \prec X\})$ and $X \subset A \cap B$.

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Determining W

If there is a constituent "*[D NP]*$_i$ *VP*" in the context, then W_i must be the set of smallest witnesses of $[\![\text{D}]\!]\,([\![\text{NP}]\!])([\![\text{VP}]\!])$

Partee's contrast

(59) Nine of the ten marbles have been found ...

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a. ...The last marble has not yet been found.

b. ...The last marble has yet to be found.

c. ...The missing marble is still unaccounted for.

d. ...Correct. The tenth marble is still missing.

e. ...The final marble is still missing.

(GPT-3 `text-davinci-003`: temperature = 0.7)

Partee's contrast

(60) a.# Nine of the ten marbles have been found.
We're still looking for it.

b. One of the ten marbles is still missing.
We're still looking for it.

Partee's contrast

GPT agrees with the judgment:

(61) Nine of the ten marbles have been found.
We're still looking for …

a. …one more.

b. …the tenth marble.

c. …the tenth marble.

d. …the tenth marble.

e. …the tenth marble.

(GPT-3 `text-davinci-003`: temperature = 0.7)

Partee's contrast

GPT agrees with the judgment:

(62) One of the ten marbles is still missing.
We're still looking for …

a. …it.

b. …it.

c. …it.

d. …it.

e. …it.

(GPT-3 `text-davinci-003`: temperature = 0.7)

Partee's contrast

Partee's contrast serves as a cautionary tale against too liberal a theory of pronoun accessibility.

Partee's contrast

Here, the two sentences license different W, predicting the contrast.

(63) a. Nine of the ten marbles$_1$ have been found.

b. $\rightsquigarrow W_1 = \{X \mid |X| = 9 \wedge \text{marbles}(X) \wedge \text{found}(X)\}$

(64) a. One of the ten marbles$_2$ is still missing.

b. $\rightsquigarrow W_2 = \{X \mid |X| = 1 \wedge \text{marbles}(X) \wedge \text{missing}(X)\}$

Interim summary

- What referents a quantifier makes available can be systematically derived from its truth-conditions.
- Specifically, a quantified statement allows one to refer to the individuals which bear witness to its truth.

Determining f

Pronoun structure

(65) $[\![\text{it}_i]\!] = f_i(W_i)$

If there is just one witness, different choice functions will yield the same truth-conditions: $f_i(\{p\}) = p$

(66) There is a phone-book and it is in the cabinet.

What if there are multiple phone-books?

(67) There is a phone-book$_i$ and it$_i$ is in the cabinet.

Is the sentence true if:

- there are two phone-books
- one is in the cabinet, the other isn't ?

What if there are multiple phone-books?

(67) There is a phone-book$_i$ and it$_i$ is in the cabinet.

Is the sentence true if:

- there are two phone-books
- one is in the cabinet, the other isn't ?

$\rightsquigarrow$ *intuitions are not sharp!*

A vexed question:

- [Mandelkern, 2020]: *"This is a complicated issue, which I won't explore in detail here."*
- [Elliott, 2020b] : *"How to capture existential vs. universal readings is a thorny issue, and [...] therefore we leave a more thorough exploration of donkey anaphora [...] to future work."*

$\rightsquigarrow$ likewise, my answer will be incomplete!

(68) There is a phone-book$_{37}$ and it$_{37}$ is in the cabinet.

Three hypothetical readings have been discussed:

- $\forall$-reading : *every phone-book that there is is in the cabinet*
- $\exists$-reading : *at least one phone-book is in the cabinet*
- uniqueness reading: *there is just one phone-book and it's in the cabinet*

Which readings obtain?

Note: while typically discussed for donkey sentences [Kanazawa, 1994], the question is raised in *all* environments.

(69) a. If there is a phone-book, it is in the cabinet.

b. Either there isn't a phone-book or it's in the cabinet.

c. Every person that has a phone-book puts it in their cabinet.

What we know about the readings of anaphora

The presence of a default reading has been confirmed experimentally many times in donkey sentences. [Foppolo, 2008, Denić and Sudo, 2022, Sun et al., 2020]

(70) a. Some farmer who has a donkey feeds it oats.
$\rightsquigarrow$ *feeds one of their donkeys oats* ($\exists$, 90%)

b. Every farmer who has a donkey feeds it oats.
$\rightsquigarrow$ *feeds all of their donkeys oats* ($\forall$, 60%)

c. No person who has a donkey feeds it oats.
$\rightsquigarrow$ *feeds any of their donkeys oats* ($\exists$, 90%)

What we know about the readings of anaphora

But certain lexical items are biased and can override the "*default*" reading.

(71) a. No person who had an umbrella left it at home today.
$\rightsquigarrow$ *left all of their umbellas in the cabinet* ($\forall$)

b. Jane has an umbrella and she left it at home.
[Chatain, 2018]

There remain some knowledge gaps:

- What about other environments (*and*, bathroom sentences)?
- What about intuitions of uniqueness?
- Are these truly readings?

There remain some knowledge gaps:

- **What about other environments (*and*, bathroom sentences)?**
- **What about intuitions of uniqueness?**
- Are these truly readings?

$\rightsquigarrow$ *joint work with Benjamin Spector and Nina Gregorio*

Generalizing to the worst case, we try to derive all possible readings, not accounting for "*default readings*".

Choice of f

A sentence S containing pronouns is:

- definitely true if true for all choices of f
- definitely false if false for all choices of f
- undetermined otherwise

Generalizing to the worst case, we try to derive all possible readings, not accounting for "*default readings*".

Choice of f

A sentence S containing pronouns is:

- definitely true if true for all choices of f
- definitely false if false for all choices of f
- undetermined otherwise

Depending on the question under discussion and other pragmatic factors, undetermined may resolve to true or may resolve to false [Kriz, 2015, Champollion et al., 2017].

(72) Every farmer who has a donkey$_1$ feeds it$_1$ oats.
where $W_1(g)$ = the set of donkeys owned by $g(x)$

a. **Definitely true:**
for all f,
every farmer who has a donkey
feeds f(donkey of $g(x)$) oats

b. **Definitely false:**
for no f,
every farmer who has a donkey
feeds f(donkey of $g(x)$) oats

This determines the following truth-/falsity-conditions.

(73) Every farmer who has a donkey$_1$ feeds it$_1$ oats.

a. ...feeds all of them oats (true, $\forall$)

b. ...feeds some but not all of them oats (undetermined)

c. ...feeds none of them oats (false, $\neg\exists$)

Depending on the pragmatics, the reading can oscillate between $\exists$ and $\forall$.

Conclusion

Summary

- With the proposed theory, pronoun accessibility fully reduce to presupposition satisfaction.

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- With the proposed theory, pronoun accessibility fully reduce to presupposition satisfaction.
- No need for enrichment ; the semantics is kept mostly intact.
- **Predictions:**
  - Bathroom sentences.
  - Donkey, modal subordination.
  - Availability and unavailability of cataphora.
  - Intrusion of pragmatic inferences in pronoun accessibility.

Extensions

- The theory offers a different view on what needs to be acquired.
- One's knowledge of the proper use of pronouns is dependent on one's knowledge of the proper use of presuppositions.
- In acquisition, a parallel development of both would be expected.

Thank you!

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